

SCIENCE TO GOAL

Deutscher Fußball-Bund (German Football Association)

Cognitive Enhancement of Football Performance

The Role of NeuroTracker for Athlete Assessment and Conditioning

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Abstract

Sports like football require players to process massive amounts of information under high mental and physical pressure. Complex decisions must be made rapidly, while suppressing distractions and trying to make sense of unpredictable environments. Decades of sports science research has shown that cognitive abilities are a defining characteristic of elite athletic performance. However, there has been a scarcity of scientific evidence regarding practical ways to train cognitive abilities that transfer to improved athletic performance.

NeuroTracker is a new cognitive training paradigm which uses a combination of stereoscopic 3D, multiple object tracking, speed thresholds, and a wide field of view, in order to train top-down mental abilities. This training methodology shows promise for assessing and improving athletic performance. The concepts and practical applications of NeuroTracker are discussed, and four key studies are presented.

The first two of these studies indicate a useful profiling role for NeuroTracker measures; clearly differentiating elite athletes from amateur and non-athletes, as well as predicting the on-court performance of NBA players across a season. The third study demonstrates that NeuroTracker training can robustly provide near transfer to a wide range of high-level cognitive functions, known to be relevant to athletic performance. The fourth study provides evidence that NeuroTracker training can transfer to competitive football (soccer) performance, with significant improvements in passing decision-making accuracy.

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